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MILITARY SPECIFICATION

MOBILE TEST FACILITY, GAS TURBINE ENGINE TEST SYSTEMS

GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes engineering and design criteria for the development and construction of NAVAIR Mobile Facility (MF) Units. These MF's or units are class 1 nonexpandable tactical shelters used for testing as defined in MIL-STD-907. The nominal exterior dimensions are; 8-feet wide, 8-feet high and 20 feet long in the shipping mode configuration. The Mobile Testing Facility shall be referred to hereafter as the MF or unit.

1.2 International Interest. Certain provisions of this unit are the subject of an international standardization agreement, ISO 1496/1.

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, Systems Engineering and Standardization Department. (Code 53), Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2330

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* 2.1 Government documents. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.).

SPECIFICATIONS

Federal

FF-P-101	Padlock
TT-E-529	Enamel, Alkyd, Semi-gloss
ZZ-R-765	Rubber, Silicone

MILITARY

MIL-P-514	Plate, Identification, Instruction and Marking, blank
MIL-F-5591	Fasteners, Panel, Nonstructural
MIL-A-9067	Adhesive Bonding, Process and Inspection, Requirements for
MIL-W-6858	Welding, Resistance, Spot and Seam
MIL-F-14072	Finished for Ground Electronic Equipment
MIL-N-18307	Nomenclature and Identification for Aeronautical Systems including Joint Electronics type Designated Systems and Associated Support Equipment.
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-M-43719	Marking Materials and Markers Adhesive, Elastomeric, Pigmented, General Specification For
MIL-C-83286	Coating, Urethane, Aliphatic, ISO Cyanate, for Aerospace Applications

STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification, Marking of U.S. Military Property
MIL-STD-401	Sandwich Constructions and Core Materials; General Test Methods
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification and Processes for Corrosion Prevention
MIL-STD-808	Finish, Materials, and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-907	Engineering And Design Criteria For Shelters, Expandable And Non-Expandable

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(Unless otherwise indicated, copies of federal and military specifications and standards, are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia,Pa. 19111-5094

2.2. Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues ofthe documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY for TESTING and MATERIALS (ASTM)

ASTM C423-89 Standard Test for Sound Absorption of Acoustical Materials in Reverberation Rooms

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute

ANSI MH5.1.1 Closed Van Containers, Requirements For

(Applications for copies should be addressed to the American National Standard Institute, 1430 Broadway, New York, N.Y. 10018)

International Organization for Standardization (ISO)

ISO 1496/1 Freight Containers-Specification and Testing

(Application for copies should be addressed to the International Organization for Standardization (ISO), 1 rue de Varembe, 1211 Geneva 20, Switzerland

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except the related associated detail specifications, specification sheet, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 6.3.) in accordance with 4.4. Unless otherwise specified, the manufacturer shall submit two units for first article testing. The units submitted shall be representative of the construction, workmanship, components and materials to be used during production. Fabrication of additional units prior to approval of the first article samples will be at the contractor's risk, and such additional items are subject to rejection in the event they fail to meet the requirements of this specification. After first article approval, one of

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the units submitted shall be delivered as directed. The second unit shall be retained at the contractor's plant to be used as a manufacturing standard and will be presented for inspection as the last unit for delivery under the contract.

3.2 Specification sheets - The mobile facility requirements shall be as specified herein and in accordance with applicable specification sheets. In the event of conflict between the requirements of the basic specification and the specification sheet, the requirements of the specification sheet shall prevail.

3.3 Size - The unit size shall be nonimally 8-feet wide, 8-feet high, and 20-feet long and shall conform specifically to dimensions and tolerances as establish in the American National Standards Institute (ANSI) ANSI MH5.1.1 and the size and tolerance specified in the applicable specification sheet.

3.4 Weight ratings - The weight rating on which the structural design and testing of the unit is based shall be in accordance with ANSI MH5.1.1 and the applicable specification sheet.

3.5 Construction - The unit described in this specification and the applicable specification sheet shall be designed, constructed and tested in accordance with this specification, and ANSI MH 5.1.1 except that all design requirements for handling and securing for commercial air mode are waived.

3.5.1 Materials and components - Materials and components shall conform to the applicable specifications, and be as specified herein. Materials which are not covered by specifications or standards shall be of the best commercial quality, the lightest practical weight, and entirely suitable for the purpose intended.

3.5.1.1 Corrosion resistance - There shall be no degradation of shelter components due to fungus growth. All materials including hardware and fastening devices shall be either of corrosion resistant material or processed to be corrosion resistant.

3.5.1.2 Fungus resistance - The unit and all materials therein shall be resistant to fungus. Materials shall be selected to maximize inhibition of fungus growth.

3.5.1.3 Dissimilar materials - The use of dissimilar metals used in e contact with each other shall be in accordance with MIL-STD-889. Galvanic corrosion shall be avoided by using an insulated material to minimize the corrosive effect.

3.5.1.4 Protective treatment - Materials that are subject to deteriortion, when exposed to environmental conditions likely to occur during service usage, shall be protected against deterioration in a manner that will comply with the performance requirements of this specification.

3.5.2 Panels.- Each wall, roof and floor of the unit shall be constructed of two essentially parallel aluminum skins enclosing an insulating core material and required structural members. The roof floor panel skins shall be made of one piece seamless construction. The side and end panels may be seamed only above and in line with the sides of the door cutout. The insulating core material shall be incombustible or self-extinguishing polyurethane foam slab and shall be bonded to both the inner and outer skins without voids or delaminated areas. The bonding

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material shall be flexible and allow for slight movement. Nothing shall protrude from the inside surfaces of the unit except items such as door latches, handles and International Standards Organization (ISO) corner fittings. All interior surfaces of each panel shall be flat within 1/8 inch and intersecting panels mutually perpendicular to within 1/16 inch when measured in accordance with this specification.

3.5.2.1 Mounting members.- The design of the walls, roof and floor mounting members shall permit direct installation of a closed end, watertight threaded insert which accommodates a 5/16-18 UNC bolt from the inside of the unit. All lateral mounting members shall be located on 22-inch centers. longitudinal members, if used, shall extend the full length of the unit and shall be located to meet the strength requirements of this specification.

3.5.2.2 Thermal barrier - A thermal barrier shall be bonded between all mounting members and the inside skin. Rubber shall not be used for this purpose.

3.5.2.3 Panel core material bond - Adhesive bonding of the panel core material shall be in accordance with MIL-A-9067.

3.5.2.4 Adhesive - The adhesive used to make structural connections in the unit shall meet the test requirements of this specification.

3.5.2.5 Vapor seal - The inner and outer skins and all joints shall provide a vapor barrier against the entrance of moisture to the core material and to the interior of the unit. The vapor barrier shall be maintained intact at all cross-sectional openings of the walls, floor and roof. All joints and edges shall be assembled and treated in a manner to prevent collection and retention of moisture. All rivets, screws and fasteners shall be dipped in sealer prior to installation. The sealing compound shall meet the test requirements of this specification. The vapor seal shall exhibit sufficient flexibility to allow movement of the outer panels due to vibrations or temperature variations.

3.5.3 Floor - The floor of the unit shall be capable of withstanding a uniformly distributed load of not less than the design gross weight of the shelter.

3.5.4 Roof - The roof of the unit shall be capable of withstanding a uniformly distributed load of not less than 660 pounds on an area of 24 inches x 12 inches.

3.5.5 Impact resistance. - All MF panels shall be designed to withstand the impact test requirements of this specification.

3.5.6 Hardware - Hardware for the unit consisting of such items as door handles and cover fasteners shall be operable by personnel wearing arctic-type gloves. The large mittened hand is defined as extending the human hand size 1/2 inch in all directions. Screws and fasteners which retain access devices (doors, covers, plates, panels and butting kits, etc.) shall be captivated and shall have slotted or cross recessed heads. Screws shall be capable of complete, individual loosening prior to movement of the access device. Quarter-turn fasteners may be used only to retain non-structural access devices, where quick access is necessary. The fasteners shall conform to MIL-F-5591. No hardware shall protrude beyond the dimensions established in ANSI MH5.1.1.

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3.5.7 Corner fittings - Top and bottom corner fittings conforming to the dimensional and structual requirements of ANSI MH5.1.1 shall be provided as an integral feature of each corner Of the structural member.

3.5.8 Forklift capability - The unit shall be designed to permit movement by a forklift truck. Such movement shall be accomplished from either side of the unit without jeopardizing its structural integrity. Four heavy-duty, reinforced forklift pockets, dimensioned and located in accordance with ANSI MH5.1.1 shall be provided within the floor area of the unit. The bottom side of each pocket may be left open; a minimum of five retainers, of sufficient strength, shall be provided across the bottom to prevent the unit from tipping off the forklift tines.

3.5.9 Doors - All doors specified in the applicable specification sheet shall be readily removable when in the open position. The design of the door shall be such as to preclude removal when the door is in a closed and latched position. All doors shall be flush mounted wherever possible, but in no case may any part or structural member extend beyond the dimensions established by ANSI MH5.1.1. The doors shall be fabricated of the same materials as the wall panels and shall have complete gasketing conforming to the requirements of this specification (see 3.7.2 and 3.7.3). Gasket materials shall conform to ZZ-R-765 Class 3b, Grade 50. Each door shall be equipped with drip caps installed directly above the door. The door latching device shall be not less than a three (3) point roller latch. Assembly operating at the top, center and bottom of the door. The roller latch mechanism shall be actuated simultaneously by handles provided on the exterior and interior of the door. The door shall be capable of being locked from inside or outside of the unit. A heavy duty padlock conforming to FF-P-101, Type F shall be provided and securely fastened by chain to outside of the unit. Provision shall be made for a method of emergency operation of the door from the inside when the outside is padlocked. A device shall be provided to securely hold the door when it is opened to the maximum position, approximately, 180°.

3.5.9.1 Louvers - Each personnel door shall be equipped with rainproof, light-proof, dustproof louvers with dust filters, hinged and sealed covers to provide proper ventilation. The louvers shall be installed as high as practical in the door and shall be 6 inches wide by 6 inches high.

3.5.10 Jacks - When specified (see paragraph 6.2) in the contract or order, removable corner jacks shall be provided by the supplier. The design selected for the jacks must have a positive lock, no-slip construction, ratchet gear drive (chain type jack head is not acceptable) with a lifting capacity of 5000 pounds. The jacks shall be constructed of high-grade, lightweight, corrosion and rust-resistant materials and shall be capable of lifting a unit whose gross weight is 20,000 pounds a minimum of 52 inches. The jacks shall be attached directly to the corner posts without the use of separate jack retaining pads. The jack mounting surface of the unit shall be compatible with existing jacks which will be provided as GFM (Government Furnished Material). The interface between the jacks and the unit shall be designed so that the weight of the unit shall be supported by the jack body and not dependent on the shearing strength of the attaching bolts. All tapped holes in the unit for the mounting bolts shall be rustproof.

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3.5.11 Slings - When specified (see paragraph 6.2) in the contract or order slings shall be provided for lifting the unit; however, at least one sling assembly shall be provided for test purposes under each contract or order. The sling assembly shall consist of a lifting ring and four legs. Each leg shall be equipped with a 12 inch turnbuckle and two safety hooks. One hook shall be compatible with the corner fittings without the use of shackles. The sling assembly shall be designed and constructed to pass the tests of this specification.

3.5.12 Butting kit - When specified (see paragraph 6.2) in the contract or order, butting kits shall be provided in accordance with the applicable specification sheet.

3.5.13 Corner fitting engaging and securing devices - When specified (see paragraph 6.2) in the contract or order, the contractor shall select and provide any of the following devices. These devices shall be designed and constructed to be compatible with the corner fittings of the unit. The loads for the securing devices are defined in ANSI MH5.1.1. The securing devices are as follows;

a. Adjustable End-To-End Couplers

b. Twist Lock Stacking Fittings

c. Twist Lock Tie-Down Fittings(with corner guidance)

d. Deck Securing Pads(open base type)

3.6 Design - The Units procured under this specification and the applicable specification sheet shall have the following design features.

3.6.1 Stacking - The unit shall meet the stacking requirements of ANSI MH51.1

3.6.2 Lifting from the top - The units shall meet the lifting from top requirements of ANSI MH51.1.

3.6.3 Restraint - The unit shall meet the restraint requirements of ANSI MH5.1.1.

3.6.4 Rigidity - The unit shall meet the rigidity requirements of International Standard ISO 1496/1.

* 3.6.5 Electric Power Distribution - The primary electric power distribution shall be designed and installed in accordance with the applicable specification sheet.

3.7 Service conditions - The unit shall meet the following service conditions.

3.7.1 Ambient temperature range - Ambient temperature in the range of -40°F to +125°F plus a solar load such that the outside skin reaches a minimum temperature of +180°F.

3.7.2 Waterproofness - The unit shall be completely waterproof.

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3.7.3 Overall Coefficient of heat transfer - The overall coefficient of heat transfer of the unit, with all access openings closed, shall not exceed 0.25 British Thermal Units (BTU) per hour, per square foot, per degree Fahrenheit.

3.7.4 Moisture resistance - The unit shall be capable of withstanding five continuous 48-hour cycles of the moisture resistance test of MIL-STD-170.

* 3.7.5 Front Panel air conditioning - The front panel and the slide assembly for the air conditioned unit shall be in accordance with the corresponding Engine Test System Drawings.

3.8 Interchangeability - Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the unit.

3.9 Markings -

3.9.1 Interior marking - Each wall, roof and floor mounting member centerline shall be marked on the inside surface of the unit.

3.9.1.1 Instruction plate - An instruction plate shall be provided on each personnel door indicating the emergency exit procedures. This plate shall be in accordance with the specification sheet.

3.9.2 Exterior markings - Exterior markings including warnings and cautions, decals, and data plates shall be in accordance with the NAVAIR drawing package for each MF type.

3.9 2.1 Louver marking.- Above each louver which is installed in the personnel doors and in the solid ends of SOMFs B and C type, stencil the following;

"OPEN LOUVER DURING AIR TRANSPORTATION"

3.9.3 Identification - Parts and assemblies shall be identified in accordance with MIL-STD-130. An identification plate shall be provided in accordance with MIL-N-18307 and shall contain the following information;

Manufacturer's name, code and model number:
Contract number:
Delivery date:
Navy serial number:
Stock number: (Block to be left blank)
Technical manual number:
Government inspector (stamp):
Type equipment code:

The identification plate shall be located in accordance with the applicable specification sheet.

3.9.4 Navy serial number - The serial number shall be in accordance with MIL-M-43719, Type I, Class 1. The letters and numeral shall be three inches high and the color shall be black. The serial number shall be located approximately seven inches down and seven inches from the left end on each side of the unit and centered near the top of each end of the unit

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3.9.5 Transportation data plate - A transportation data plate shall be provided in accordance with MIL-P-514 and located adjacent to identification plate. The plate shall contain the following information;

Shipping Weight: (Block to be left blank)
Shipping Cubage:
Overall Length:
Overall Width:
Overall Height:
C.G. Location: (Block to be left blank)
Permissible Modes of Transportation: Military Air, Marine and highway

3.9.6 Convention on safe containers (CSC) safety approval plate - A CSC Safety Approval Plate shall be provided in accordance with the international Convention for the Safe Containers Act of 1977 (PL95-208-DEC 13, 1977 and 49- CFR450-453). The plate shall be 4 inches high by 8 inches wide, permentaly affixed to the MF and contain the following data fields:

Plate Serial No.
Date Manufactured
Maximum Gross Weight in KG and LBS
Allowable Stacking Weight in KG and LBS
Racking Test Load Value in KG and LBS
End Wall Strength
Side Wall Strength
Month and Year of Inspection

3.10 Finish - Finish of the unit shall be in accordance with the applicable specification sheet.

3.11 Skids - Skids shall be in accordance with the applicable specification sheet.

3.12 Workmanship - All parts, components and assemblies of the unit including castings, forgings, molded parts, stampings, seals and sealing agents, machined surfaces and welded parts shall be clean and free from any defects that will reduce the capability of the unit to meet the requirements specified herein. External surfaces shall be free from burrs slag and sharp edges.

3.12.1 Welding - Spot welds shall be in accordance with MIL-W-6858. Surfaces to be welded shall be free from rust, scale, paint, grease and other foreign matter. All welds shall be of ample size, with thorough penetration and good fusion and shall be free of scabs, blisters, pockmark, bruises or other surface defects. Welded surfaces shall be thoroughly cleaned and any excuse flux shall be chemically neutralized and removed. Welders shall be qualified in accordance with MIL-STD-248.

3.12.2 Riveting - Riveting shall be in accordance with best commercial practice. Countersunk holes for flush rivets shall not be less than the height of the rivet head. All rivets shall be tight and properly seated.

3.12.3 Castings - Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance - All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection - The supplier shall be responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions - Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in (applicable test method document or applicable paragraph(s) in the specification).

4.4 First article inspection - First article inspection shall consist of all inspections and tests of this specification and the applicable specification sheet. Only one of the two units submitted in accordance with paragraph 3.1 shall be subjected to these tests. There shall be no distortion, deformation, delamination, deterioration, malfunction or any other damage to the unit as a result of the first article testing. Failure of the first article sample to pass one or more of these tests shall be cause for rejection.

4.4.1 Failures - Failure of component(s), defects in material construction, design or non-compliance with any portion of this specification noted during construction or tests shall be cause for rejection of the unit. Subject to approval of the Contracting Officer, the contractor shall, at his cost, correct the deficiencies and resubmit the unit for retest.

4.5 Quality conformance inspection - Quality conformance inspection shall consist of individual and sampling inspection.

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4.5.1 Individual - Each unit produced under contract or order shall be examined for the following defects list in Table 1. One or more defects shall be cause for rejection.

Table 1. Classification of Defects.

Item	Description of Defect	Crit	Maj	Min
a.	Transportation data plate or Identification plate omitted, incorrect or illegible.		X	
b.	Finishes not as specified, wrong color, gloss, adhesion, or wrong type		X	
c.	Doors do not open, close, or latch when operated by one man.		X	
d.	Gaskets not securely fastened or missing.			X
e.	Components missing or not as specified.		X	
f.	Welded joints cracked or porous.	X		
g.	Butting kits, ECU panel omitted.			X
h.	Markings omitted, incorrect, or illegible.			X
i.	Fasteners omitted or damaged.			X
j.	Padlock or keys omitted.			X
k.	Louver filter omitted.			X

4.5.2 Sampling - One (1) sample unit out of each seven (7) shall be selected at random by the cognizant Government Inspector and tested in accordance with paragraphs 4.6.3.3 (excluding the test requirements of 4.6.2), 4.6.7 and 4.6.8. If one sample is found not acceptable the next three units shall be considered samples and be subjected to the failed test. If one or more of the three units from the second sample are found not acceptable, the lot of seven units shall be rejected, corrective action taken, and the balance of the lot tested for the failed condition.

4.6 Tests -

4.6.1 Test conditions - Unless otherwise specified, test shall be conducted at an ambient temperature of 70°±18°F and at an ambient barometric pressure of 28 to 32 inches of mercury.

4.6.2 Performance tests -

4.6.2.1 Stacking - The unit shall be subject to the stacking test ,Test no.1, of ANSI MH5.1.1.

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4.6.2.2 Lifting from the top - The unit shall be subject to the lifting from the top test ,Test no.2, of ANSI MH5.1.1.

4.6.2.3 Restraint - The unit shall be subject to the restraint test ,Test no.4, of ANSI MH5.1.1.

4.6.2.4 Rigidity - The unit shall be subject to the rigidity tests of International Standard ISO 1496/1, Test no. 9,7,10.

4.6.2.5 Forklift - The unit shall be tested in accordance with the forklift test ,Test no.13, of ANSI MH5.1.1.

4.6.2.6 Roof - The roof of the unit shall be tested in accordance with Test no. 10, of ANSI MH5.1.1.

4.6.2.7 Mounting members - A closed end, watertight, threaded insert to accommodate a 5/16-18 UNC bolt shall be installed in the mounting member of the floor and side wall from the inside of the unit. The insert, mounting member and panel shall withstand a load of (1) 2000 pounds with the load applied axially to a bolt in the insert and without the aid of bracing or support which is not part of the shelter, and (2) 40 inch pound torque load. After removal of the loads there shall be no distortion of the insert, mounting member or panel.

4.6.2.8 Door - Each door of the unit shall be subjected to a 200 pound static load applied with the door open and located at a maximum distance from the hinges for a period of at least 30 minutes. The door shall show no signs of deformation or malfunction. The device designed for emergency operation of the door from the inside when the outside is padlocked shall be demonstrated.

4.6.2.9 Jacks - When jacks are a requirement of the contract or order (see paragraph 6.2) they shall be tested in accordance with the following;

a. Four jacks shall be attached to a unit loaded to a gross weight of 20,000 pounds and raised simultaneously to a height of 30 inches then lowered to the starting position.

b. Twenty lifting cycles shall be conducted during one of which the unit shall be raised to a height of 52 inches.

There shall be no evidence of damage or deformation to any part of the jacks or the unit.

4.6.2.10 Sling assembly test - When sling assembly are a requirement of the contract or order they shall be tested as follows;

a. Two lifting rings shall be subjected to a destructive tensile load test.

The two lifting rings shall show no signs of material or mechanical failure at 60,000 psi.

b. Three legs of a sling assembly shall be subjected to a destructive tensile load test.

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The three legs shall show no signs of material or mechanical failure at 20,000 psi.

c. A unit loaded to a gross weight of 30,000 lbs shall be supported by the sling assembly for not less than thirty minutes.

There shall be no sign of damage or deformation to the unit or the sling during the test.

4.6.2.11 Impact -

4.6.2.11 Panel surfaces - All panel surfaces shall be tested to impact resistance test requirements. A 24 inch square specimen of each of the wall, floor and roof construction, having the least apparent impact strength and representative of having no support from an internal structural member, shall be subjected to impact as follows;

a. A 70-pound steel cylinder, three inches in diameter and hemispherical spherical at one end, shall be dropped vertically 30 inches so that the that the hemispherical end of the weight strikes the outer skin of the section on a horizontal plane. For this test the specimen shall be supported along its four edges by a frame 3-5/8 inches high, 24 inches square outside, made from lumber 1-5/8 inches thick.

After the impact test there shall be no fracture or rupture of either of the skins and the inner skin shall not be dented.

4.6.2.12 Roof - A sling assembly shall be attached to the four upper corner fittings and extended to its maximum height. The sling assembly shall be dropped in free fall so that the assembly strikes the roof of the shelter. The roof shall experience no fracture or rupture at the completion of the test.

4.6.2.13 Sound Absorption - The unit shall be tested for sound absorption in accordance with ASTM C423.

4.6.2.14 Riveting.- The riveting shall meet all test and visual inspection requirements.

4.6.2.15 Lamination.- Panels and the MF shall be inspected and tested for delamination. The unit shall pass all inspection and test requirements.

4.6.3 Environmental tests - The unit under test shall not sustain any permanent deformation, delamination, core material change, buckling, corrosion or other deterioration and there shall be no malfunction of the doors, latches, hinges or covers as a result of the following tests.

4.6.3.1 Ambient Temperature.- The unit shall be tested in accordance with the extreme-temperature cycle of MIL-STD-810. The exposure at +125°F shall be for at least 12 hours. During this 12-hour period a solar load shall be applied for 4 hours to provide an outside skin temperature of at least 180°F. The exposure at -40°F shall be for 24 hours.

4.6.3.2 Moisture resistance - The unit shall be subjected to five continuous 48 hour cycles of the moisture resistance test of MIL-STD-810

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4.6.3.3 Waterproofness - The unit shall be tested for waterproofness in accordance with Test no. 15 of ANSI MH5.1.1. This test shall be conducted following the completion of all of the performance tests prescribed herein. After the completion of this test, the unit shall be completely free from any penetration of water.

4.6.3.4 Overall coefficient of heat transfer - The unit shall be placed in a temperature controlled room and all of its openings sealed with tape to prevent circulation of air between the outside and the inside of the unit. The test shall be conducted by placing the unit in a constant temperature enviroment (-40 °F) with an electrical heat source contained within the unit to prevent a temperature differential of not less than 100 °F between the interior and exterior of the unit. The heat source shall have suffecient power to maintain a stablized temperature of 100 °F sbove the ambient temperature outside the unit. The electrical power input shall equal the heat loss for the temperature differential and shall be the basis for calculating the heat transfer coefficient which shall be no greater than 0.25 BTU/Sq.Ft./°F.

4.6.3.5 Thermal shock - The largest panel used in the construction of the unit shall be used for this test. The panel shall be stabilized at 75°±5°F before being placed in a chamber maintained at a temperature of -40°±5°F. The panel shall remain in the chamber for a period of 2 hours and then quickly subjected to 75° ±5°F conditions for 2 hours. During the test a 2-inch-thick layer of core material shall be placed in contact with the exposed surface of the inside of the panel.

4.6.3.6 Fire resistance.- Panel specimens shall be tested for fire resistance in accordance with the fire resistance test of MIL-STD-907

4.6.3.7 Sunshine (Ultraviolet effects).- The unit shall be tested to determine the deterioration of external components in accordance with the solar load test (assembled shelter) of MIL-STD-907.

4.6.3.8 Marine atmosphere.- Marine atmosphere testing on fasteners and exposed hardware shall be completed in accordance with MIL-STD-810, method, 509.1 except that the salt water solution shall have a concentration of 10 percent and the exposure period shall be 96 hours. The test items shall show no evidence of corrosion or degration upon completion of the test.

4.6.4 Core material.- A sample specimen of core material measuring one foot square by three inches thick shall be weighed to within 0.01 pound and then subject to 98 percent relative humidity for a period of 120 hours. At the completion of this test the core material shall be reweighed and shall have not gained more than 0.1 pound of water.

4.6.4.1 Shear strength.- Core samples shall be tested in accordance with MIL-STD-401. The core material shall have a minimum shear strength of 10 psi. The bonding area between the panel skins and the core material shall have a shear strength greater than or equal to the shear strength of the core material.

4.6.4.2 Water absorption.- With the doors open the unit shall be subjected to MIL-STD-810 moisture resistance test, method 507.1, procedure II excluding steps 2,3,4,7, and 8. After the cycling has been completed, there shall be no evidence of delamination, cracking, corrosion, or deterioration of any part of the shelter.

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4.6.5 Adhesive .- A tensile pull test shall be performed on a sample lap joint consisting of two 1-inch strips of aluminum overlapped one-half inch and bonded with the adhesive material over no more than the specified one-half square inch area. Ten specimens shall be used for each condition and shall be tested in accorcance with the following to destruction.

4.6.5.1 Adhesive temperature shear.-Temperature of the samples shall be stablized at a specified level for at least 10 minutes as indicated by a located at the bond area. The minimum metal to metal shear strength of the fully cured adhesive and its structural bond shall be as follows;

Temperature	Shear Strength
-25°F	1500 psi
Ambient	1500 psi
200°F	400 psi

4.6.5.2 Adhesive humidity shear - Sample lap joints shall be exposed to 100 percent relative humidity at 200°F for two weeks. Then the temperature of the samples shall be stablized for at least 10 minutes. The minimum metal to metal shear strength of the adhesive shall be 500 psi.

4.6.5.3 Adhesive salt fog exposure.- Sample lap joints shall be subjected to the salt fog test of MIL-STD-810 for a period of 2 weeks using a 20 percent NaCl solution stablized at at least 90°F. The minimum metal to metal shear strengthy of the adhesive shall be at least 1000 psi.

4.6.6 Sealer - A tensile pull test shall be performed on sample joints consisting of two 1-inch strips of aluminum overlapped one half inch giving a contact area of one-half square inch and bonded with the sealer material over no more than the specified one-half square inch area. Ten specimens shall be used for each condition and shall be tested in accordance with the following to destruction.

4.6.6 1 Sealer temperature shear.- The temperature of the samples shall be stabilized at a specific level for at least 10 minutes as indicated by a thermocouple lacated at the bond area. The minimum shear strength of the fully cured sealer and its bond to the panel facing shall be as follows;

Temperature	Shear Strength
-25°F	200 psi
75°F	200 psi
200°F	75 psi

4.6.6.2 Sealer humidity shear - The sample lap joints shall be exposed to 100 percent relative humidity at least at 200°F for a period of two weeks. Then the temperature of the samples shall be stablized for at least 10 minutes. The minimum shear strength of the sealer shall be 75 psi.

4.6.6.3 Sealer salt fog exposure.- The sample lap joints shall be subjected to the salt fog test of MIL-STD-810 for a period of two weeks using a 20 percent NaCl solution stablized at 90°F. The minimum shear strength of the sealer shall be at least 200 psi.

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4.6.7 Maximum weight - The empty unit shall be weighed to determine compliance with the maximum tare weight specified in the applicable specification sheet.

4.6.8 Overall dimensions - The unit shall be measured to determine compliance with the interior and exterior dimension requirements of ANSI MH5.1.1.

4.6.9 Panel flatness - In a given wall, roof or floor panel, the inside panel surface over the mounting members and surfaces between the mounting members shall be flat 1/8 inch when measured with a 48 inch long, and flat within .005 inch, straightedge. Two 1/16 inch shims having the same thickness within .003 of an inch shall be used and located at the extremes of the edge. One measurement along the center of each mounting member and three measurements (top, middle and bottom) across each pair of mounting members shall be taken. Five or more gaps, in the lot of measurements taken, in excess of 1/8 inch shall be cause for rejection of the unit.

4.6.9.1 Panel squareness - In any two intersecting panels, the inside panel surface at each mounting member in one panel shall be mutually perpendicular to the inside panel surface at the mounting member in the other panel within 1/16 inch when measured with a 36 inch square whose two edges are perpendicular to each other within .005 of an inch. Two 1/16 inch shims having the same thickness within .003 of an inch shall be used along one edge of the square and measurements made along the other edge. The shims shall be located so that both shims touch the panel surface at the mounting member areas. Measurements shall be taken at each mounting member location and at the top, middle, and bottom of each of the four vertical corners. Ten or more gaps, in the lot of measurements taken, in excess of 1/16 inch shall be cause for rejection of the unit.

4.6.10 Interchangeability - The dimensions of doors, location of door hinges, filter openings, skids and butting kits (when butting kits are a requirement of the applicable specification sheet) shall be measured to determine compliance with the interchangeability requirements of 3.9.

4.7 Examination of preparation for delivery - Each unit shall be inspected for compliance with Section 5 of this specification.

5. PACKAGING.

5.1 General - Unless otherwise specified in the contract or order, the manufacturer shall pack for domestic shipment in such a manner as to insure acceptance by the carrying medium and safe delivery at destination.

5.2 Marking - Destination markings for shipment from the supplier's plant to the initial receiving activity shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use - The mobile facility testing-shelters are standard rigid walled tactical shelters constructured to ANSI/ISO specifications that may be configured for special field use as Inter/intraservice Tactical Support Facilities for testing and instrumentation of jet engine test cells.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. Quantity of jacks required (see 3.5.10).
d. Quantity of butting kits required (see 3.5.12).
e. Quantity and type of corner fittings engaging and securing devices required (see 3.5.13).

* 6.3 Data Requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Referenced Paragraph	DID Number	DID Title	Suggested Tailoring
4.6	DI-T-2072/ DI-MISC-80653	Test Reports	Wherever possible, use the contractor's standard test report format.

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

* 6.4 First Article - When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Standard sample - For first article testing the standard sample shall consist of 2 units (see 3.1).

6.6 Government furnished property - The contracting officer may arrange to furnish the property essential to testing and delivery of the mobile facility testing-shelter. All property to be furnished by the Government as part of the specification shall be listed and identified by part or indentifying number, or stock number. The quantity of each item required for one complete unit shall be listed. Each item entry shall be numbered in order to provide ready reference.

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6.7 Government-loan property - The contracting officer may arrange to loan the property essential to testing and delivery of the mobile facility testing-shelter. Property that the Government loans to the contractor for testing or any other purpose shall not lose its identity.

6.8 Subject term (key word) listing -

Field, Test, Aircraft Engines

Instrumentation, Jet Engine, Test

Jet Engines, Field, Test, Van

Test, Van, Jet Engines

* 6.9 International interest - Certain provisions of this specification paragraph 3.6.4. are the subject of international standardization agreement, International standard ISO 1496/1. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

* 6.10 Changes from previous issue - The margins of this specification are marked with asterisks to indicate where changes (additions, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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